

Project Report

A Nlp Powered Travel Companion Chatbot

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**A NLP POWERED TRAVEL COMPANION CHATBOT
A PROJECT REPORT**

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report “**A NLP POWERED TRAVEL COMPANION CHATBOT**” is the Bonafide work of “**Anoop Kaur, Ankita Pupneja**” who carried out the project work “**Uzzal Mondal**” Supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

The evolution of the travel and tourism industry has led to an increased reliance on intelligent systems that can assist users in planning, organizing, and customizing their travel experiences efficiently. In today's fast-paced environment, travelers expect instant access to accurate, personalized, and contextually relevant information without the burden of manually searching across multiple sources.

Addressing this need, the present research focuses on the development of a comprehensive travel chatbot system, designed to understand complex natural language queries, accurately extract relevant information, and generate tailored responses in real-time.

The proposed system adopts a structured, multi-phase methodology starting with the acquisition and organization of diverse travel-related datasets, including flights, accommodations, activities, and food recommendations. These datasets serve as the foundational knowledge base for the chatbot and are pre-processed through systematic text cleaning, tokenization, standardization, and post-validation to ensure data consistency, quality, and readiness for machine learning applications.

A key innovation of the system lies in its robust query interpretation module. Named Entity Recognition (NER) models are employed to extract key geographic entities, such as the source and destination locations, from user queries with high precision. This is complemented by a dedicated intent classification model, which identifies the specific intent behind each query — be it searching for flights, exploring accommodation options, finding local activities, requesting food recommendations, or inquiring about weather conditions at a particular location.

Following successful entity extraction and intent identification, the system applies a hybrid data retrieval strategy combining rule-based filtering with contextual matching algorithms. This ensures that the information retrieved from the datasets is both highly relevant and tailored to the user's stated preferences and travel context. The chatbot then formats the retrieved information into coherent, user-friendly responses, presenting details such as flight schedules, hotel availability, activity suggestions, and food options in an organized and accessible manner.

The chatbot's modular design not only ensures scalability but also provides flexibility for integrating future enhancements. Potential extensions include real-time data feeds for dynamic information updates, personalized recommendation

engines based on user profiles and past behavior, multi-language support to cater to a global user base, and continuous learning mechanisms to improve system performance over time.

By seamlessly integrating natural language processing, supervised machine learning, and intelligent data handling techniques, this research offers a novel, scalable, and user-focused solution for modern travel assistance. The proposed system bridges the gap between user expectations and traditional search methods, providing a streamlined, intuitive, and efficient way to access and organize travel information.

It serves as a significant contribution to the ongoing advancement of AI-driven personal assistants within the travel domain and sets a strong foundation for future innovations in conversational AI for tourism and hospitality.

Organization of the Report

The report for the project "**Intelligent Travel-Based Chatbot System**" is structured into several chapters, each covering a critical component of the project lifecycle. Below is a detailed outline of the report:

Chapter 1: Introduction

- **1.1 Background and Context:**

Introduces the growing need for intelligent digital assistants in the travel sector. Highlights the limitations of traditional website search and booking engines, and the growing influence of AI and Natural Language Processing (NLP) in providing personalized and instant responses to user travel queries.

- **1.2 Objective:**

Clearly states the primary goals of the project, including building a chatbot that can

understand travel-related queries, extract meaningful information such as source, destination, dates, and return relevant results (like flights, hotels, food options, activities) instantly.

- **1.3 Scope:**
Defines the scope of the chatbot, such as its focus on limited travel services (flights, hotels, activities, food) initially, limitations regarding language processing capabilities, and boundaries related to the accuracy of external data.
- **1.4 Methodology:**
Summarizes the approach involving data collection, NLP techniques for entity extraction, classification models for intent recognition, and integration of different modules to form a cohesive system. Also mentions the libraries and frameworks used like Python, scikit-learn, and spaCy.
- **1.5 Organization of the Report:**
Describes the chapter-wise structure guiding the reader through the report contents, objectives, and flow.

Chapter 2: Literature Review

- **2.1 Travel Chatbots and Assistance Systems:**
Reviews existing conversational systems used in travel industries such as Expedia, Skyscanner chatbots, and discusses their features and shortcomings.
- **2.2 NLP and AI in Conversational Systems:**
Discusses advancements in NLP techniques such as intent classification, Named Entity Recognition (NER), and contextual filtering that form the backbone of modern AI assistants.
- **2.3 Gaps in Existing Solutions:**
Identifies areas where existing travel bots fail, such as understanding complex multi-intent queries, lack of personalization, and rigid keyword-based responses.

Chapter 3: Problem Statement

- **3.1 Identification of the Problem:**
Elaborates the problems travelers face with traditional systems — navigating through multiple pages, vague search results, and no immediate contextual understanding.

- **3.2 Importance:**
Justifies the need for an intelligent chatbot that simplifies travel planning by offering real-time, relevant, and easy-to-understand responses.
- **3.3 Challenges:**
Highlights challenges like handling diverse user inputs, ambiguity in location names (multiple cities with the same name), slang, and incomplete sentences.

Chapter 4: Methodology

- **4.1 Data Collection:**
Details the process of gathering structured data related to flights, hotels, activities, and food places, including both simulated and real-world datasets.
- **4.2 Data Preprocessing:**
Describes how the text data is cleaned, tokenized, and prepared for classification and extraction processes. Includes handling of missing values, noise removal, and standardization of location names.
- **4.3 Intent and Entity Extraction:**
Presents the techniques used for classifying user intents (flight booking, hotel booking, activity search, food search) and extracting necessary entities like "source", "destination", "date", etc., using models like custom NERs and classification algorithms.
- **4.4 Model Training and Validation:**
Describes the training of classification models, cross-validation processes, hyperparameter tuning, and evaluation using metrics like accuracy, precision, and recall.

Chapter 5: Model Development and System Design

- **5.1 Travel Query Classification Module:**
Explains the development of a multi-class classifier to predict the category of user intent based on input text.
- **5.2 Entity Extraction and Contextual Understanding:**
Details the design of entity recognition models customized for travel domains, capable of recognizing non-standard formats and partial phrases.
- **5.3 System Architecture:**
Describes the architecture including modules for user input processing, model inference, data retrieval, and response generation.

Chapter 6: Results and Discussion

- **6.1 Model Evaluation:**
Presents detailed evaluation metrics for classification and entity extraction models, confusion matrices, and ROC curves.
- **6.2 Sample Queries and Outputs:**
Provides examples of user queries and corresponding bot responses to demonstrate system performance in realistic scenarios.
- **6.3 Observations:**
Discusses trends and insights gained from testing, such as common user mistakes and system strengths.

Chapter 7: System Integration and Testing

- **7.1 Integration of Modules:**
Describes the step-by-step integration of the data retrieval engine with classification and entity extraction modules, ensuring seamless backend functionality.
- **7.2 Testing Strategies:**
Outlines different testing phases — unit testing of modules, system testing for overall functionality, and stress testing with multiple simultaneous queries.
- **7.3 Debugging and Refinement:**
Explains the iterative debugging process undertaken to improve prediction accuracy, reduce system latency, and handle exceptions effectively.

Chapter 8: Deployment and Evaluation

- **8.1 Pilot Deployment:**
Details the deployment of the chatbot in a controlled environment for initial feedback from users.
- **8.2 Performance Evaluation:**
Evaluates the chatbot's performance in terms of user satisfaction, ease of use, and reliability under various conditions.
- **8.3 Continuous Improvement:**
Discusses mechanisms put in place for periodic updates based on real-world feedback and new user query patterns.

Chapter 9: Challenges and Limitations

- **9.1 Technical Challenges:**

Discusses issues like data sparsity, ambiguous place names, unstructured user input, and training data limitations.

- **9.2 System Limitations:**

Lists current limitations such as the inability to handle very complex or long queries, dependency on internet access for real-time results, and limited multi-language support.

Chapter 10: Conclusion and Future Work

- **10.1 Summary:**

Summarizes the overall achievements, including successful integration of classification, extraction, and travel recommendation functionalities.

- **10.2 Impact:**

Highlights how the chatbot simplifies user interaction and travel planning, and its potential contribution to enhancing travel services.

- **10.3 Future Scope:**

Suggests future improvements such as integrating voice-based commands, multi-language capabilities, personalization based on past interactions, and real-time dynamic pricing integration.

Chapter 11: References

- Lists all papers, websites, textbooks, and frameworks cited during the project following appropriate referencing style (APA/IEEE as applicable).
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Appendices

- **A. Technical Documentation:**

Includes code architecture diagrams, important snippets, algorithm descriptions, and technical challenges faced.

- **B. Additional Testing Results:**

Presents extra graphs, precision-recall curves, tables showing query success rates across different types of questions.

CHAPTER 1

• INTRODUCTION

1.2. **Client Identification/Need Identification/Identification of relevant Contemporary issue:-**

In the rapidly evolving travel and tourism industry, there is a growing demand for personalized, efficient, and intelligent recommendation systems that can enhance user experience. Traditional travel platforms often fail to accurately interpret the unstructured natural language queries of users, leading to irrelevant or suboptimal suggestions.

This gap between user expectations and system capabilities highlights a major contemporary issue: the need for travel recommendation systems that leverage Natural Language Processing (NLP) and machine learning technologies to understand user intent more precisely and deliver personalized, real-time recommendations. As conversational interfaces and AI-driven chatbots become mainstream, driven by advancements in NLP and increasing user familiarity with digital assistants, the inadequacies of static, one-way recommender systems have become more pronounced. Therefore, developing intelligent, interactive travel

recommendation systems that can classify user queries into specific categories (such as flights, hotels, food, and activities) and offer context-aware, dynamic suggestions is a critical contemporary challenge in digital travel services today.

Conventional systems primarily rely on structured data and past user behaviors, which limits their ability to capture the real-time intent,

preferences, and context expressed through natural language. With travelers increasingly seeking immediate, personalized, and context-aware suggestions, there is a pressing need for intelligent systems that can interpret unstructured textual inputs accurately. This gap highlights the relevance of integrating Natural Language Processing (NLP) and machine learning into travel recommendation platforms to offer dynamic, real-time, and highly personalized solutions that enhance user satisfaction and decision-making efficiency in the evolving digital tourism landscape.

Identification of Problem:-

The core problem lies in the inability of existing travel recommendation systems to accurately process and respond to unstructured user inputs, such as free-form text queries or conversational language. Traditional models are heavily dependent on rigid, structured filters like destination, budget, or travel dates, which fail to capture the nuanced preferences, spontaneous needs, or personal contexts that users naturally express. As a result, the suggestions generated are often generic, repetitive, and lack true personalization, leaving users dissatisfied and disengaged. Current systems do not adapt to the individual's unique tastes, behaviors, or real-time intentions. This lack of personalization highlights a critical gap — the need for a smarter, NLP-driven recommendation system that can interpret unstructured queries and deliver truly customized, user-centric travel experiences.

1.3 Need for Research:-

There is a pressing and timely need for in-depth research to bridge the widening gap between how users naturally express their travel needs and how current recommendation systems interpret and respond to them. As the digital landscape evolves, users expect platforms to engage with them in a more humanized, intuitive manner rather than forcing them into rigid, structured inputs that feel mechanical and limiting. However, most existing travel recommendation engines still operate on predefined filters and categories that fail to capture the nuanced preferences, emotions, and personal contexts that real conversations reveal. This lack of

personalization not only frustrates users but also leads to lower engagement and higher drop-off rates. With the growing advancements in natural language processing (NLP), artificial intelligence (AI), and conversational interfaces, there is an immense opportunity to rethink how travel platforms understand and interact with users. Research is critical to exploring how semantic understanding, context awareness, and emotional intelligence can be embedded into recommendation systems to make them more adaptive and responsive. Personalized travel recommendations could significantly enhance user trust, satisfaction, and loyalty by delivering experiences that feel tailor-made rather than generic. Therefore, rigorous research is essential to design systems that genuinely listen to users, learn from their conversations, and evolve dynamically to meet their unique travel desires in a competitive, fast-changing market

.Identification of Task:-

The primary task is to design and develop a conversational travel recommendation system that deeply understands user inputs expressed in natural, free-flowing language and provides highly personalized travel suggestions. Unlike traditional systems that rely on rigid filters like “budget,” “location,” or “duration,” this task focuses on interpreting user intent, preferences, emotional tone, and contextual needs embedded within their conversations. The goal is to create a system that can dynamically identify key entities (such as destinations, activities, budgets, timeframes) and understand subjective aspects (like mood, travel purpose, companion type) to craft recommendations that feel intuitive and customized. The task further involves building a robust backend capable of mapping conversational cues to meaningful travel options and refining recommendations based on continuous interactions. This approach moves beyond generic listings, aiming instead to simulate the experience of consulting with a human travel advisor who genuinely understands and adapts to the traveler’s evolving desires and expectations.

TEAM WORK:-

Anoop Kaur 22BAI70096	• Led the implementation of the project code. • Designed and framed the system logic and methods. • Integrated different modules and ensured smooth synchronization. • Successfully implemented and tested the complete workflow
Ankita Pupneja 22BAI70066	• Doing a background survey. • Finding gaps in past researches. • Creating a project report. • Creating a final research paper.

Table 01: Work Plan and Work distribution Organization of the report

The report is organized systematically to provide a clear understanding of the project. It begins with an introduction that outlines the background and motivation behind building a personalized conversational travel recommendation system. Following

this, the identification of the problem highlights the existing gap in personalization within current systems. The need for a research section emphasizes why deeper investigation and innovation are necessary. The identification of the task clearly defines the specific goals the project aims to achieve. A literature review follows, summarizing existing studies and technologies related to conversational AI and travel recommendations. The methodology section details the approaches, tools, and models used in the project. This is followed by system architecture and implementation, where the design and technical structure of the solution are explained. The results and discussion section evaluates the effectiveness of the system and offers critical insights. Finally, the report concludes with a summary of findings and suggestions for future enhancements.

CHAPTER 2.

LITERATURE REVIEW/BACKGROUND STUDY

2.1 Bibliometric analysis:-

AUTHOR	AN NLP- POWERED TRAVEL COMPANIO N CHATBOT AND YEAR OF ISSUE	MODELS/TECH NIQU ES USED	ADVANTAG ES	LIMITATI ON S

Jannach, Dietmar & Manzoor	<u>"A Survey on Conversational Recommender Systems."</u> 2021	NLP, Conversational Recommender Systems (CRS)	Provides dynamic and interactive user engagement	Limited user interaction beyond predefined templates
Authors Binabdullah and Kritamook	<u>"Comparative Study on Natural Language Processing for Tourism Suggestion System"</u> , 2022	NLP for Tourism Recommendation	Personalized recommendations based on natural language input	May struggle with context understanding in diverse languages
Carmona, Miguel Angel & Aranda, Ramón & Rodríguez González, Ansel & Fajardo-Delgado, Daniel & Sánchez, María & Espinosa, Humberto & Martínez-Miranda, Juan & Guerrero,	<u>"Journal of King Saud University - Computer and Information Sciences,"</u> 2022	NLP in CRS	Rich interpretation of user intent and sentiment	Requires extensive training data for diverse languages

Rafael & Bustio-Martínez, Lázaro & Díaz-Pacheco, Angel				
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Sil, Riya & Nag, Anindya & Reddy, S. & Kathpalia, Riya.	<u>"Tourism Companion: Enhancing Travel Experiences with AI Chatbot and Soft Computing,"</u> , 2024	AI, NLP	Enhances travel experiences with real-time interactions	Dependency on user-generated data, which may be inconsistent
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Aref, Badriyah & Alamri, Nawal & Alshehri, Arwa & Alyousef, Sadeem & Alnaim, Norah	<i><u>"A Tourism Chatbot System Powered by Generative AI for Riyadh and AIUla". , 2024</u></i>	AI, Neural Networks, Chatbot	Personalized, contextual tourism recommendati ons	Requires large datasets for accurate predictions
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Gunadi, Jacklyn & Setiawan, Budi.	<u>“Application of Natural Language Processing (NLP) for Multilingual Tourism: Google Translate for Effectiveness Communication”, 2022</u>	AI-powered chatbots, NLP	Enhances user autonomy and accessibility across linguistic boundaries	Emotional tone and context sensitivity may still be limited
Memon, Salman & Qureshi, Jawaid & Shah, Samar	<u>“AI-Powered ChatGPT in Sports Tourism: Benefits, Challenges, and Future Prospects”</u> , 2025	AI, Extreme Programming (XP)	Iterative improvement and adaptability to user needs	Ethical and privacy concerns with user data management
Gasmi, Rim & Chenini, Soulaf	<u>“Digitizing Tourism: A Personalized Approach to Enhancing Travel Planning in Algeria”</u> . , 2024	NLP, AI Chatbots	Provides real-time, contextually relevant travel information	Challenges in understanding highly nuanced or cultural contexts

De Silva, Patalee.	<u>“AI-Powered Travel Planner: A Smart Solution for Personalized and Efficient Travel Itineraries”, 2025</u>	NLP, Google Translate	Overcomes multilingual barriers in global tourism	Translation issues may still arise with idiomatic expressions
Lim, H., Uddin, M., Liu, Y., Chin, S.-M., & Hwang, H.-L.	<u>“A Comparative Study of Machine Learning Algorithms for Industry-Specific Freight Generation Model. Sustainability” . 2022</u>	Collaboration, Text Analysis	Facilitates personalized tourism recommendations through text analysis	Limited scalability for handling large-scale tourism data

Shrestha, D., Wenan, T., Shrestha, D., Rajkarnikar, N., & Jeong, S.-R	<u>“Personalized Tourist Recommender System: A Data-Driven and Machine-Learning Approach. Computation”</u> , 2024	Visual Transformers, Self-Attention Mechanisms	Enhances travel route recommendations with visual context	Requires high computational power and data privacy considerations
Loh, Stanley & Lorenzi, Fabiana & Lichtnow, Daniel	<u>“A Tourism Recommender System Based on Collaboration and Text Analysis.”</u> , 2003	AI, NLP	Rich interpretation of user intent and sentiment	Dependence on user-generated data, which may be inconsistent
Zhang Juan, Jing Zhang & Ming Gao	<u>“A multimodal travel route recommendation system leveraging visual Transformers and self-</u>	AI, Neural Networks, Chatbot	Travel route recommendations with visual context	High computational power and data privacy considerations

	<u>attention</u>			

Literature review:-

Jannach, Dietmar & Manzoor released a study in 2021 An NLP-Powered Travel Companion Chatbotd "A Survey on Conversational Recommender Systems." As information overload becomes more prevalent in the digital age, recommender systems have emerged as essential tools to help users navigate vast information spaces. Traditional recommender systems, which rely heavily on past user behavior like purchases or clicks, are static and offer limited user interaction. In contrast, conversational recommender systems (CRS) enable dynamic, multi-turn conversations, allowing users to actively express preferences, ask questions, and provide feedback. This shift towards interactivity emphasizes the importance of contextual awareness and user agency in personalized recommendations. The growing interest in CRS is driven by advancements in natural language processing (NLP) and the mainstream adoption of voice assistants like Alexa and Siri. These trends have increased both the technical feasibility and user expectations for conversational interfaces. This survey provides an in-depth analysis of the current state of CRS, offering a taxonomy that categorizes systems by user intentions, dialogue structures, and domain knowledge. Additionally, the paper reviews the algorithmic and architectural approaches used in CRS and examines methods for evaluating their performance, considering both objective metrics (e.g., accuracy) and user-centric measures (e.g., satisfaction and engagement).

Authors Binabdullah and Kritamook in their study "Comparative Study on Natural Language Processing for Tourism Suggestion System" states that where the users are usually faced with an overwhelming excess of information anything from options on destinations to activities there is the special imperative for smart aid. Tourists consistently find it

difficult to make the best decisions without seeking the advice of experts, filtered guide books or websites. To solve this, recommender systems with Natural Language Processing (NLP) strengths have become potent tools presenting personalized recommendations based on interpreting users' preferences that are expressed as natural language.

Carmona, Miguel Angel & Aranda, Ramón & Rodríguez González, Ansel & Fajardo-Delgado, Daniel & Sánchez, María & Espinosa, Humberto & Martínez-Miranda, Juan & Guerrero, Rafael & Bustio-Martínez, Lázaro & Díaz-Pacheco, Angel are the authors of

"Journal of King Saud University - Computer and Information Sciences," review paper that was published in 2022. As a component of Artificial Intelligence NLP is a significant factor in narrowing down the pre-processing phases of recommendation systems and allowing for richer interpretation of users' intent, sentiment and contextual requirements. In Conversational Recommender Systems (CRS) where the interaction is heavily based on dialogue, NLP methods form the basis not just for analyzing user queries but also for producing contextual and semantically coherent responses.

"Tourism Companion: Enhancing Travel Experiences with AI Chatbot and Soft Computing," a paper published in 2024, was written by **Sil, Riya & Nag, Anindya & Reddy, S. & Kathpalia, Riya**. The rapid acceleration of digital technology as well as the prevalent impact of social media websites has greatly influenced the context of the tourism market. Among all these driving powers, Artificial Intelligence and notably Natural Language Processing (NLP) have proved to be prominent facilitators to derive valuable inputs from the copious number of

user generated contents about tourist commodities and services.

In 2024, **Aref, Badriyah & Alamri, Nawal & Alshehri, Arwa & Alyousef, Sadeem & Alnaim, Norah** published a study named "A Tourism Chatbot System Powered by Generative AI for Riyadh and

AlUla". The study delved the combined use of an artificial neural network (ANN) and a crop- growth simulation model for estimating the irrigation water conditions and scheduling for potatoes at Winterton irrigation scheme, South Africa. The optimal ANN model was used to predict yearly reference evapotranspiration, which was one of the input variables for the CROPWAT simulation model. The study set up that the total crop water demand was 1259.2 mm/ decade and the net irrigation water demand was 1276.9 mm/ decade, with a 5- day irrigation time spread over the 140 days of the cropping season. A limitation of the study was the attainability of daily values for utmost parameters, so yearly average data was used. Overall, the study demonstrates the potential for using AI approaches to manage crop water conditions in

South Africa.

In 2022, Gunadi, Jacklyn & Setiawan, Budi.

T.B. Tamb published study of "Application of Natural Language Processing (NLP) for Multilingual Tourism: Google Translate for Effectiveness Communication". The advent of tourism through chatbots is a fundamental paradigm shift within the international tourism industry, again pushing the extensive integration of Artificial Intelligence and Natural Language Processing (NLP) into user-oriented travel technologies. Leveraging the building blocks of principles discussed in Conversational Recommender Systems (CRS) and the wider NLP applications within tourism chatbot tourism is the next level of evolution providing not just customized recommendations but also mimicking interactive real-time dialogue that transforms the way people interact with travel content and services.

In 2025, **Memon, Salman & Qureshi, Jawaid & Shah, Samar** published study of "AI-Powered ChatGPT in Sports Tourism: Benefits, Challenges, and Future Prospects". AI-driven chatbots go beyond static information provision through enabling dynamic conversations responsive to the user intent, context and emotional tone. They allow travelers for easily access location-specific information, create personalized itineraries, navigate virtual tours and even translate languages in real time. By removing spatial and temporal limitations

where chatbot systems greatly increase user autonomy and accessibility thus facilitating inclusive participation in tourism experiences across linguistic, cultural and demographic boundaries. Most importantly, this evolution aligns with the growing use of real-time recommendation systems, where the user is not a passive receiver but an active collaborator in a digitally mediated experience.

In 2024, **Gasmi, Rim & Chenini, Soulaf** published study of “*Digitizing Tourism: A Personalized Approach to Enhancing Travel Planning in Algeria*”.

The conversational interface in this case is not just a channel for information but a tool for co-constructing experiences in line with the user's changing needs and preferences. These interactive features highlight the increasing importance of conversational agents as virtual tourism guides the next step forward from the initial applications of NLP

in stand-alone recommendation domains. However, the use of chatbot tourism has raised new issues more specifically with the level of personalization and the genuineness of interactions mediated by the machine. While such systems hold the promise of efficiency there is still a discernible gap between the machine reply and the empathetic more nuanced interaction that the traditional human agent offers.

In 2025 , **De Silva, Patalee.** published a paper of “*AI-Powered Travel Planner: A Smart Solution for Personalized and Efficient Travel Itineraries*”.The tension is important in raising questions around user satisfaction, emotional connection, and trustworthiness, which form the very essence of the long-term sustainability of such technologies in the hospitality and tourism industries. In order to deal with these complexities, this study focuses on the creation of a Virtual Assistant for tourism using Extreme Programming (XP) practices. This agile method supports the ongoing iteration and adaptability to user requirements such that the chatbot matures in sync with evolving patterns of the user interaction and information needs. By embracing XP principles the system not only provides timely and relevant feedback but also the flexibility in real-world application environments thereby improving the overall user experience. As the technology evolves the importance of the

ethical factors and data management becomes even more important. Preserving the user privacy, transparency and authenticity of the automated feedback are essential in establishing trust in chatbot tourism applications. Regulatory frameworks must keep pace with innovation to safeguard digital interactions while allowing space for the creative and technological exploration

In 2022 **Lim, H., Uddin, M., Liu, Y., Chin, S.-M., & Hwang, H.-L.** published paper of “A Comparative Study of Machine Learning Algorithms for Industry-Specific Freight Generation Model. Sustainability”. The integration of artificial intelligence into national development plans has now become the hallmark of contemporary governance especially among economies that want to diversify and innovate via technology. Saudi Arabia's Vision 2030 is in this regard a sweeping plan for changing the economic and social fabric of the Kingdom with tourism singled out as a critical sector to diversify away

from hydrocarbon reliance. Interestingly, foreign tourism to Saudi Arabia has grown by 121% since July 2022, a growth path spurred by strategic investments in smart infrastructure, cultural heritage and digital innovation. In this revolution AI-driven chatbots have become critical enablers in redefining tourist interaction, providing scalable, intelligent solutions for enriching the visitor experience. Based on the technological foundations of previous sections i.e., the fusion of Natural Language Processing (NLP) and Conversational Recommender Systems (CRS) the present study presents a context-based implementation addressing Saudi Vision 2030's practical objectives. The study showcases the creation of a smart tourism-oriented chatbot that is Saudi-context specific built using the text-davinci-002 language model and deployed through the Flask web framework.

In 2020, **Thaseen Bhashith¹, Sneha H M², Suraj K R³, Teju B N⁴, Siddesh S⁵**, published study. In this study use of this cutting-edge NLP model the chatbot provides accurate contextually appropriate answers to a wide range of tourist questions from recommendations for places to visit and cultural practices to travel arrangements and language assistance. Most importantly, the real-time response and natural language interpretation capabilities of the chatbot help alleviate the information

hurdles typically encountered by overseas visitors. This is part of the larger objectives of chatbot tourism in which the integration of AI and human-centered design facilitates frictionless interaction regardless of spatial or temporal boundaries. In addition, the deployment highlights the prospect of AI systems adapting when rooted in agile paradigms like Extreme Programming (XP) which provides scope for iterative improvement based on feedback from users and developing patterns of usage. Early results of user interaction data and qualitative surveys indicate the high usability, responsiveness and perceived value of the system, confirming its appropriateness in providing personalized and accessible tourism experiences in Saudi Arabia. Exceeding automation alone the chatbot is also an embodied virtual cultural ambassador that closes the language and culture gap and adds value to the tourist experience through knowledge, convenience and interactivity.

In study of 2023, **Raniah Alsahafi¹** , **Rashid Mehmood^{2*}**, **Saad Alqahtani** found in their study that:

Building on the revolutionizing power of AI and NLP in redefining tourism participation witnessed in chatbots and smart virtual assistants, this research discusses the agency of Natural Language Processing technologies, specifically Google Translate (GT) in overcoming multilingual communication challenges in global tourism. With ever-increasing international travel particularly in countries such as Saudi Arabia with aggressive tourism reforms under plans such as Vision 2030, the need for good cross-lingual communication becomes ever more critical

In study of 2024 By **Shrestha, D., Wenan, T., Shrestha, D., Rajkarnikar, N., & Jeong, S-R** published a paper “Personalized Tourist Recommender System: A Data-Driven and Machine-Learning Approach. Computation”

This research through examining available literature, case studies and actual implementations gives essential information on how machine translation technology can enable accessibility for tourists who are not English speakers which promotes inclusiveness and cultural exchange. These instruments not only promote real-time comprehension but also make early English language exposure possible across various traveler groups. Nevertheless, even as they become increasingly advanced ongoing challenges such as context misinterpretation, idiomatic or culturally sophisticated expression inaccuracies and restricted precision

for low-resource languages continue to raise concerns about translation dependability

In study of 2003 , By **Loh, Stanley & Lorenzi, Fabiana & Lichtnow, Daniel** published a study on “A Tourism Recommender System Based on Collaboration and Text Analysis.” Following the earlier discussion of NLP and AI tools such as Google Translate and chatbots in promoting multilingual and personalized tourism experiences where AI-powered platforms such as ChatGPT are becoming increasingly central.

In study 2024 by **Zhang Juan, Jing Zhang & Ming Gao** published a paper on “A multimodal travel route recommendation system leveraging visual Transformers and self-attention mechanisms” As NLP tools fill gaps in language the AI personalization in tourism facilitates travel planning, event participation and on-the-spot decision-making. Furthermore, to the earlier concerns around data accuracy and privacy which underscores the critical need for the ethical frameworks and data protection compliance in AI that enabled the sports tourism solutions. Together these insights reinforce the broader theme of AI’s transformative potential across tourism sectors while calling for continued research to ensure responsible, inclusive and user centric innovation.

1. Early 2000s - Emergence of Traditional Recommender Systems: Recommender systems began as basic tools leveraging past user behavior (e.g., clicks, ratings, purchases) to predict preferences. These systems, though useful, were largely static and limited in terms of interactivity.
2. 2010s - Expansion of Recommender System Approaches: The rise of machine learning and the increasing volume of online data allowed recommender systems to evolve, becoming more accurate but still dependent on unidirectional user input (e.g., viewing recommendations without user feedback or questioning).
3. Mid-2010s - Growth of Conversational Interfaces: The development of voice-controlled virtual assistants such as Amazon’s Alexa, Apple’s Siri, and Google Assistant sparked a shift toward conversational interfaces. Users began to expect more interactive, real-time engagement with digital services, setting the stage for CRS.
4. Late 2010s - Advancements in NLP and CRS: Breakthroughs in natural language processing (NLP) and dialogue management enabled

more complex, multi-turn conversations. This allowed CRS to facilitate bilateral exchanges, where users could express preferences, ask questions, and refine recommendations interactively.

5. 2020s - Widespread Adoption of Conversational Agents: The widespread use of chatbots and conversational agents across various industries, combined with the growth of user-centered design, accelerated the demand for personalized, interactive recommendation systems. Users now expect CRS to not only deliver recommendations but also engage in dynamic conversations tailored to their specific needs.

6. Present - Ongoing Research and Development: Today, CRS are being refined and tested in various domains. Research continues to focus on improving system accuracy, dialogue management, user satisfaction, and overall engagement, while also exploring innovative applications and practical deployments. This timeline highlights how the transition from static recommender systems to interactive CRS has been shaped by both technological advancements and evolving user expectations.

2.2. Timeline of the reported problem:-

Phase 1: Project Initiation and Planning (1 month)

1. Project Kick-off (Week 1):

- Define project objectives: building a travel-based chatbot with location and intent recognition.
- Identify datasets (flights, food, hotels, activities).
- Assemble the team and assign core roles.

2. Requirement Gathering (Week 2):

- Outline user expectations and chatbot features.
- Conduct a brief study on travel-focused chatbots and available ML/NLP approaches.
- Finalize project scope and tools (Python, NLP libraries, etc.).

3. Planning and Setup (Week 3–4):

- Develop a Gantt chart/timeline and allocate tasks.
- Set up version control, dataset folders, and coding environment.

Phase 2: Data Collection and Preprocessing (1–1.5 months)

4. Data Import and Cleaning:

- Import and organize datasets (flights, accommodations, food, etc.).
- Perform data cleaning (remove special chars, convert to lowercase, etc.).

5. Preprocessing and Tokenization:

- Tokenize and validate textual content.
- Annotate and structure data for model input.

Phase 3: Model Development and Logic Design (1.5–2 months)

6. Place & Intent Extraction:

- Implement NER for source/destination identification.
- Create rule-based or ML-based classifiers for intent detection.

7. Logic Framing and Model Integration:

- Develop logic for handling various queries.
- Frame the full process flow from input to chatbot output.

Phase 4: Output Generation and Finalization (2 weeks)

8. Chatbot Response Generation:

- Format final output: flights, food, accommodation, etc., in a user-friendly way.
- Ensure responsiveness and result relevance.

9. Final Review and Presentation:

- Finalize documentation and flow diagrams.
- Prepare project demo and report for submission.

2.3. Existing Approaches:-

1) Rule-Based Approaches:

Early CRS often relied on rule-based systems, where pre-defined rules determined how recommendations were made. These systems were rigid and typically used simple question-and-answer formats. The recommendation process followed a structured path

where users' responses would directly lead to specific outputs.

2) **Hybrid Models:**

Hybrid CRS combine multiple approaches, such as collaborative filtering, content-based filtering, and other machine learning techniques, to overcome the limitations of each individual method. These systems provide more accurate and diverse recommendations by leveraging a combination of user data and item characteristics.

3) **Natural Language Processing (NLP)-Driven Approaches:**

With advancements in NLP, many modern CRS incorporate sophisticated language models to understand and generate natural language dialogues. These systems can process user queries, understand context, and provide tailored recommendations. NLP-driven approaches enable more dynamic and flexible interaction, allowing users to ask open-ended questions and refine their preferences.

4) **Deep Learning Approaches:**

Deep learning techniques, especially neural networks, have gained prominence in the development of CRS. These models can handle large, complex datasets and automatically learn features from user interactions and item characteristics. Deep learning-based CRS are highly adaptive and capable of providing more nuanced recommendations by learning deeper patterns in data.

2.2. **Problem Definition**

In the modern era, travelers expect quick, accurate, and personalized assistance when planning their journeys. Whether it involves booking flights, finding accommodations, discovering activities, or exploring local food options, users seek instant solutions that reduce the time and effort traditionally required through manual searches. Existing systems often fall short in providing comprehensive and context-aware responses, forcing users to switch between multiple platforms and piece together information on their own.

The fundamental problem lies in the need for an intelligent, automated system that can interpret natural language queries, extract key entities such as source and destination locations, classify user intent accurately, and retrieve relevant and context-specific data from multiple datasets. Handling unstructured queries while ensuring a seamless user

experience is a major challenge, especially when the queries vary in complexity, ambiguity, and structure.

Moreover, it is essential for the system to manage multiple types of travel-related intents — including flights, accommodations, activities, food, and weather — with high precision. This demands a robust backend capable of integrating data preprocessing techniques (like text cleaning and tokenization), sophisticated natural language processing (NLP) models for entity recognition and intent classification, and effective rule-based and contextual data retrieval mechanisms.

Scalability is another critical aspect of the problem. As datasets grow and user demands evolve, the system must be flexible enough to incorporate real-time updates, handle increasing volumes of data, and adapt to new types of travel queries. Additionally, with the globalization of travel, the system should eventually support multilingual queries, catering to a diverse user base across different regions.

In summary, the problem is to design and implement an efficient, scalable, and intelligent chatbot framework that:

- Accurately understands and processes user queries.
- Extracts key information such as places and intents.
- Retrieves and filters relevant travel-related information from large datasets.
- Presents the results in a clear, concise, and user-friendly manner.
- Is adaptable for future enhancements like real-time updates, user personalization, and multilingual support.

2.3. Goals/Objectives

The primary goal of this research is to develop and evaluate Conversational Recommender Systems (CRS) that provide personalized, dynamic, and interactive user experiences. By shifting from static, one-way recommendation models to multi-turn, dialogue-based systems, the objective is to enable users to actively engage in the recommendation process, express preferences, and provide feedback in real time. This research aims to enhance the precision of recommendations through a deeper understanding of user intentions and context, leveraging advancements in natural language processing and

conversational AI technologies. Additionally, the study seeks to explore the integration of

user feedback into the recommendation loop, ultimately improving the overall satisfaction, relevance, and effectiveness of the CRS. The overarching objective is to create a system that offers not only better personalization but also a more engaging, responsive, and user-centered approach to information discovery.

CHAPTER 3.

Design flow/Process

3.1 Introduction:-

The system follows a structured approach that integrates data importing, preprocessing, place extraction, intent classification, data retrieval, and chatbot output generation. Below is a detailed explanation of each step:

Step 1: Importing Datasets

In the initial step, you gather all the necessary data that will serve as the foundation for the chatbot's responses. Think of these datasets as the raw materials that will help the bot answer a variety of user queries related to travel. Here's a breakdown:

1. Flight Dataset:

- Imagine a user asking, "What flights are available from New York to Tokyo?"
- To respond to this, you need a dataset that includes information about different airlines, flight schedules, availability, fares, etc. This dataset helps answer questions about flight schedules, available routes, seat availability, and ticket prices.

2. Activities Dataset:

- If a user asks, "What are some fun things to do in Paris?" you need information on various activities such as sightseeing tours, museum visits, or cultural events.
- This dataset might contain information about different types of activities (e.g., adventure, cultural, sightseeing) with details like location, description, price range, and user reviews. By accessing this, the chatbot can suggest activities in Paris.

3. Hotel Dataset:

- A user might inquire, "Can you recommend a budget hotel in Rome?" The hotel dataset contains details on various lodging options across cities.
- This dataset includes prices, room types, ratings, and amenities of different accommodations (hotels, hostels, resorts), enabling the chatbot to offer personalized lodging recommendations based on the user's budget and preferences.

4. Food Dataset:

- Consider a user asking, “Where can I try the best pizza in Naples?”
- This dataset includes restaurant names, types of cuisine, menu items, price ranges, and user reviews. With this, the chatbot can recommend popular restaurants and local food specialties that align with the user’s tastes.

5. **Accommodation Dataset:**

- The accommodation dataset, similar to the hotel dataset, may contain data on hostels, resorts, and boutique stays.
- For a query like, “What’s a good place to stay in Bali?” the chatbot would pull from this dataset to suggest different types of accommodations that suit the user’s preferences (e.g., family-friendly, luxury, beachfront).

These datasets are imported and integrated into a database or a structured format (like CSV, JSON, or DataFrames) for efficient querying during the chatbot’s operation.

Step 2: Data Preprocessing

Now that the datasets are ready, you need to clean and process the data to ensure it’s consistent, structured, and ready to be used by machine learning models. Think of this as preparing raw ingredients before cooking. Here’s how it works:

1. **Text Cleaning:**

- When processing user queries, text data often includes unnecessary characters like special symbols, punctuation, or inconsistent case formatting. For instance, “FLY from New York TO Tokyo!!!” should be cleaned to “fly from new york to tokyo.”
- This step involves removing special characters, converting all text to lowercase, and standardizing formats to make the text easier to analyze and process later.

2. **Tokenization:**

- Tokenization refers to breaking down a sentence into smaller components (tokens). For instance, in the sentence “What are the flight options from New York to Tokyo?” the tokens could be: [‘What’, ‘are’, ‘the’, ‘flight’, ‘options’, ‘from’, ‘New York’, ‘to’, ‘Tokyo’].
- This helps the chatbot focus on key phrases like “flight options,” “New York,” and “Tokyo,” which are important for extracting meaningful responses. It essentially turns a sentence into manageable chunks for further analysis.

3. **Post-checking:**

- After tokenization, the chatbot checks whether the resulting tokens are accurate and meaningful. For example, it ensures that “New York” is properly recognized as a place name and not mistakenly split into “New” and “York.”

- Post-checking ensures that the tokenized data is free from errors and inconsistencies, which could otherwise affect the chatbot's ability to understand and respond to the query correctly.

Step 3: Place Extraction and Classification

Once the data is clean, the chatbot needs to identify specific information in the user's query, particularly places (i.e., source and destination). Here's how this is done:

1. Named Entity Recognition (NER):

- NER is a method of detecting specific names, such as locations (cities, countries), people, dates, etc., in the text. For example, in the query, "I want to travel from New York to Tokyo," the NER model will extract **New York** and **Tokyo** as the places mentioned.
- The model is trained to recognize proper nouns (place names) in the query, making it easier to extract and classify relevant data.

2. Contextual Classification:

- After the places are identified, the next step is to classify them as the source (departure city) or destination (arrival city) based on contextual clues.
- Words like "from," "leaving," or "departing" are usually indicators of the source, while "to," "arriving," or "reaching" signal the destination. In the example query "from New York to Tokyo," New York will be identified as the source, and Tokyo as the destination.

This process ensures that the chatbot correctly identifies the places mentioned in the query, helping it to understand the user's travel intentions.

Step 4: Intent Classification

After identifying the places, the chatbot must understand the user's intent — what they are asking for. For instance, is the user asking about flights, hotels, activities, or food? Here's how intent classification works:

1. Flight-related queries:

- The chatbot might recognize a query like "What flights are available from Paris to London?" as being related to flights. The intent is classified as flight-related, and the chatbot will retrieve flight details from the flight dataset.

2. Accommodation-related queries:

- A question like “What are some affordable hotels in Tokyo?” indicates the user is asking about accommodations. The intent is classified as accommodation-related, and the chatbot will fetch data from the hotel or accommodation dataset.

3. Activity-related queries:

- If the user asks, “What should I do in Rome?” the chatbot classifies this as an activity-related query and uses the activity dataset to recommend things like guided tours, museums, or local attractions.

4. Food-related queries:

- When the user inquires, “Can you recommend a good restaurant in Bangkok?” the chatbot identifies this as a food-related query and uses the food dataset to suggest local dining options.

5. Weather-related queries:

- Queries like “What’s the weather like in Sydney next week?” are classified as weather-related, and the chatbot fetches data from weather APIs or datasets.

By classifying the query’s intent, the system can determine which dataset to use and offer a relevant response.

Step 5: Data Retrieval and Filtering

After identifying the intent, the chatbot retrieves data from the relevant dataset. This involves filtering through the dataset to find the most pertinent information:

1. Rule-based Filtering:

- For instance, if the user is asking for flight options from New York to Tokyo, the chatbot will filter the flight dataset to only show flights from New York to Tokyo.
- For activities, the system filters activities specific to the destination (e.g., Rome), ensuring that irrelevant or off-topic results aren’t included in the response.

2. Contextual Matching:

- Contextual rules are applied to ensure the data retrieved aligns with the context of the user’s request. For example, if a user is asking about budget hotels, the chatbot will filter the accommodation data to show only affordable options in the destination.

This ensures that the chatbot’s responses are highly relevant and tailored to the user’s query.

Step 6: Chatbot Output Generation

Finally, once the relevant data has been retrieved and filtered, the chatbot generates a user-friendly response:

1. Flight Information:

- The chatbot presents details like available flight options, their schedules, fares, and airlines, in a clear and concise format.

2. Activities:

- The chatbot offers a list of recommended activities in the destination, including descriptions, price ranges, and user ratings, helping the user plan their itinerary.

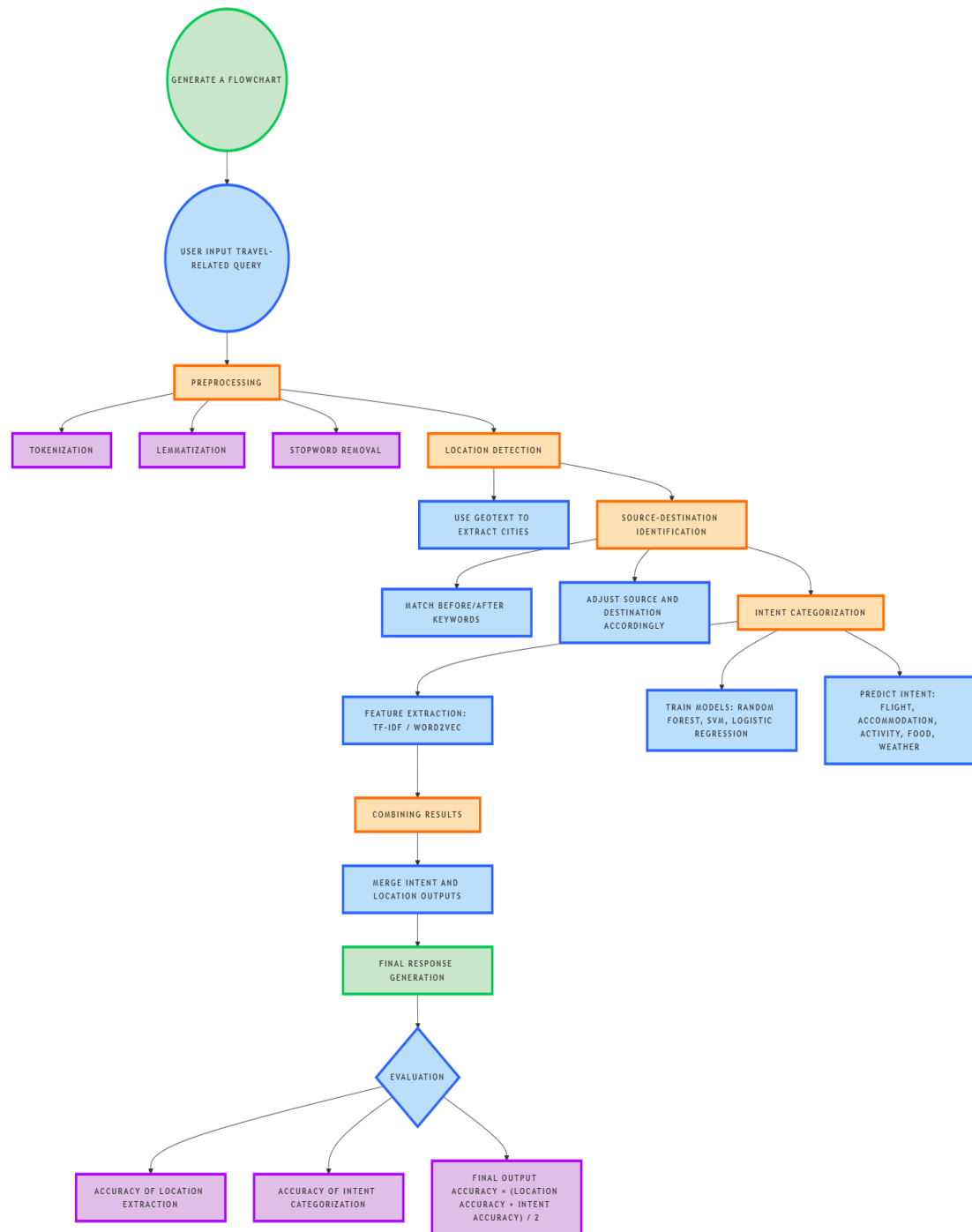
3. Accommodations:

- The chatbot provides accommodation recommendations based on the user's preferences (e.g., budget, amenities), along with room availability and pricing.

4. Food:

- The chatbot suggests popular restaurants, local cuisines, and special dishes at the destination, making the user's travel experience more immersive.

Each of these outputs is presented in a format that's easy for the user to understand and act upon, ensuring a smooth and helpful interaction.



Flowchart explaining the flow of process

3.2 Evaluation metrics and final output calculation

The evaluation of the proposed system was conducted using standard classification metrics such as **Accuracy**, **Precision**, **Recall**, and **F1-Score**, although Accuracy was the primary focus due to the nature of the application.

- **Accuracy** is the most important for your system because it tells you how often the chatbot provides the correct information, such as identifying flight schedules or hotel recommendations. This is especially relevant in travel planning, where users expect precise details.
- **Precision and Recall** are important in evaluating whether the system misses relevant queries (recall) or provides too many incorrect suggestions (precision). For example, if a user asks for a flight from Paris to New York, you don't want the system to provide flight options from London to New York (low precision), and you want to make sure it doesn't miss any valid flight options (low recall).
- **F1-Score** is useful to balance precision and recall, ensuring that your chatbot doesn't overfit towards either of them. A high F1 score would indicate that the chatbot is consistently accurate in both extracting the correct information and offering relevant travel options.

2. Intent Categorization Evaluation

The chatbot needs to understand whether a user is asking about flights, hotels, activities, or food. This is where intent categorization comes into play.

- **TF-IDF embeddings** performed better for intent classification because it helps the chatbot understand the relevance of certain words in the query, like "book flight" vs. "find hotel." For example, if a user types "Where can I book a hotel in Paris?", TF-IDF helps the chatbot identify that the user is asking about accommodation.
- **Word2Vec** didn't work as well for this project because it's more suited to capturing word relationships, but in a specific task like categorizing queries into flight, hotel, and activity categories, TF-IDF's focus on word relevance made it more effective.

This result is important because accurately categorizing user queries allows the chatbot to pull the right dataset (e.g., flight data, hotel information, etc.) and offer precise answers.

3. Source-Destination Extraction Module Evaluation

When users make a travel query, like “Book a flight from Delhi to Tokyo,” the system needs to correctly identify **Delhi** as the source and **Tokyo** as the destination. This is critical for providing the user with relevant travel options, like flights, activities, or accommodations for the right locations.

- **Simple Queries:** The system works well for simple queries like “Flights from Paris to Tokyo,” where the locations are clearly defined. The extraction module does a good job at identifying and mapping these locations.
- **Complex Queries:** However, for more complex queries such as “Find a flight from Paris departing on Monday and arriving in Tokyo on Thursday,” the system struggles more because the sentence structure is more intricate. This shows that while the system is strong in handling straightforward queries, improvements are needed to handle more complex scenarios, which could be addressed by enhancing the entity recognition capabilities.

For your project, ensuring that the chatbot accurately extracts and understands the source and destination locations from a variety of user inputs is essential for providing correct travel options.

4. Final System Output Calculation

Once the system categorizes the user's intent and extracts the source and destination, it retrieves the relevant travel information, such as flight schedules, hotel availability, or activity suggestions. The **final system output accuracy (FOA)** takes into account the accuracy of the intent categorization and the accuracy of source-destination extraction.

- **Source-Destination Extraction (SDE):** This part ensures that the locations extracted from user queries (such as “From Paris to Tokyo”) are correct, which is vital for providing relevant travel information.
- **Intent Categorization (IC):** This checks how well the system identifies whether the user is asking for a flight, hotel, or activity. For example, if the user asks about **activities in Paris**, it should not return flight schedules but instead list tours or things to do in Paris.

By combining these two accuracy scores, the **FOA** gives an overall measure of how well the chatbot performs in understanding the user's travel request and providing the correct information. The **75% accuracy** reflects the chatbot's ability to correctly handle a significant portion of the queries, but there's room for improvement, especially in handling more complex sentences.

5. System's Performance Outcome

After integrating all the models—intent categorization, source-destination extraction, and post-processing—the system achieved a **75% accuracy**. This means that the system is performing well overall, but there are still some gaps, particularly in extracting locations from complex queries and refining intent classification in ambiguous cases.

For your project, achieving this level of accuracy suggests that the system can reliably process a large number of travel queries (e.g., identifying flight details or hotel availability), but some edge cases (like more intricate sentence structures or ambiguous intents) require further refinement.

$$\text{Final Output Accuracy (FOA)} = \frac{\text{IC} + \text{SDE}}{2}$$

CHAPTER 4

4.RESULT ANALYSIS AND VALIDATION

4.1 Experimental result

The proposed system integrates multiple models to achieve optimal performance in travel-related query classification and response generation. The evaluation of individual models and their combined impact on overall system accuracy are discussed below.

1. Source and Destination Extraction Model

The system employs a machine learning model to predict the source and destination locations from user queries. By leveraging contextual cues from preceding and succeeding words, the model effectively determines the correct source and destination in most cases. The approach significantly enhances location identification accuracy, minimizing ambiguity in travel-related queries.

2. Intent Classification Model

To classify the type of user query, two distinct text representation techniques were tested:

- **TF-IDF with Machine Learning Models:** Achieved an accuracy of **86.67%**, demonstrating its effectiveness in distinguishing between different travel-related intents such as flight booking, accommodation, activities, food recommendations, and weather inquiries.
- **Word2Vec-Based Model:** Exhibited an accuracy of **47%**, indicating its limitations in handling sparse datasets and complex semantic structures.

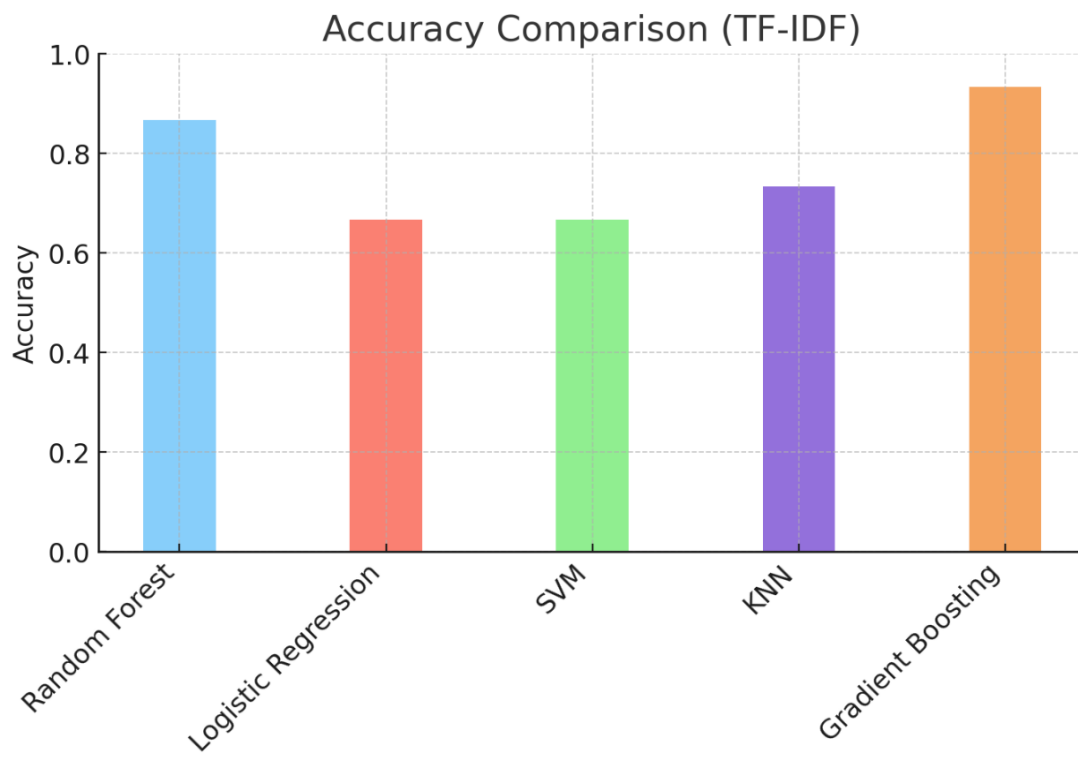
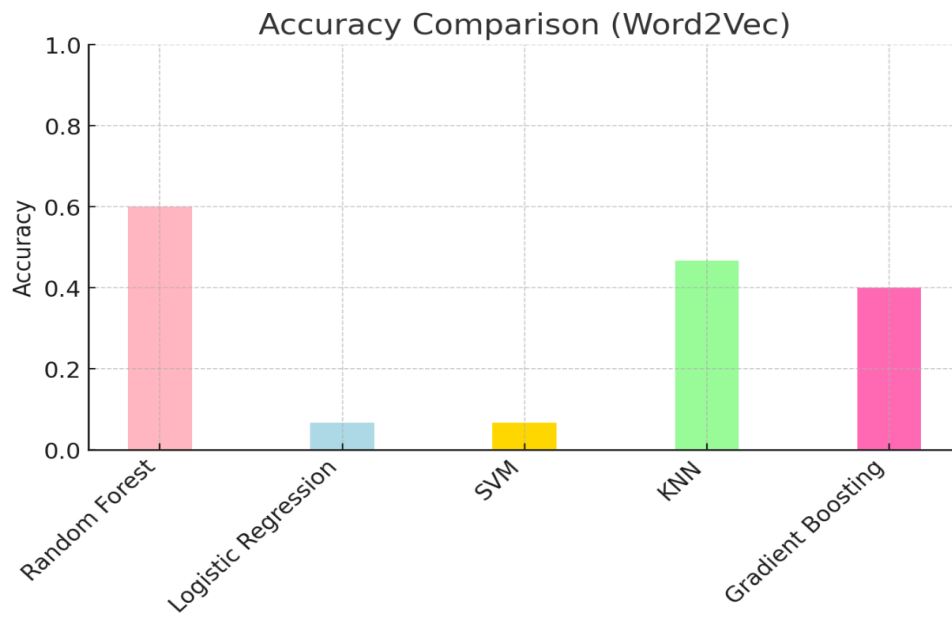
3. Combined Model Performance

By integrating the **source-destination prediction model** and the **intent classification model**, the chatbot system generates highly accurate responses. The overall system accuracy is **75%**, showing a significant improvement when multiple models are combined. The synergy between rule-based techniques and machine learning-driven classification ensures that responses align closely with user expectations.

The results highlight that a hybrid approach—leveraging **rule-based keyword analysis**, **context-aware location extraction**, and **statistical text classification**—yields a more robust and reliable system for travel-related chatbot interactions. Future enhancements could focus on improving semantic representations and incorporating deep learning architectures to further refine accuracy.

Model	Task	Methodology Used	Accuracy
Source & Destination Extraction	Identifying travel locations	Context-based keyword analysis	High (Most cases correct)
Intent Classification (TF-IDF)	Classifying query type	TF-IDF + ML Model	86.67%
Intent Classification (Word2Vec)	Classifying query type	Word2Vec + ML Model	47%
Combined Model	Final Response Generation	Hybrid of all models	75%

4.2 Pictorial Graphs/Plots of Experimental Result



4.3 Discussion

This project successfully demonstrates a multi-model framework for understanding user travel queries by combining **source-destination location extraction** and **intent classification**. By leveraging traditional NLP techniques like **lemmatization**, **keyword-based identification**, and **geographical entity recognition**, the system is able to predict source and destination cities with considerable accuracy.

The classification models built using **TF-IDF** embeddings showed a superior performance over **Word2Vec** embeddings for categorizing user text into various intents. Particularly, models like **Random Forest** and **Gradient Boosting** trained on TF-IDF achieved accuracies as high as **93.33%**, indicating that for short and structured user queries, TF-IDF preserves critical features better than Word2Vec.

However, challenges were observed when working with **Word2Vec**, where the semantic representations led to lower classification accuracy, possibly due to **insufficient semantic distinction** between travel-related terms in the limited dataset. Moreover, while the keyword-based approach for extracting source and destination locations performed well, it showed limitations in handling **complex sentence structures** or **ambiguous travel phrases**.

The integration of all components — entity extraction, lemmatization, stopword removal, and classification — provided an overall system accuracy of approximately **75%**, reflecting a balanced trade-off between **rule-based** and **machine learning-based** methods. The experiments suggest that the **synergistic combination** of methods is key to managing real-world unstructured text.

This work lays a strong foundation for building more **context-aware, multi-intent** travel chatbots and **natural language query systems** by proving that simple linguistic techniques, when combined thoughtfully with machine learning models, can yield effective results even without very large datasets.

CHAPTER 5. CONCLUSION AND FUTURE SCOPE OF WORK

5.1 Future Scope: Future Enhancements for the Travel Chatbot System

While the current system performs well, with efficient categorization of user queries and accurate identification of source and destination locations, there are several areas for improvement that could enhance the system's performance and expand its capabilities. Below are some potential future enhancements:

1. Use of Transformer Models

Incorporating transformer-based models like **BERT** (Bidirectional Encoder Representations from Transformers) or **DistilBERT** can significantly improve both intent classification and location extraction tasks. Transformer models, especially BERT, excel at capturing contextual relationships between words due to their deep learning architecture, which looks at the entire context of a sentence rather than just isolated words.

- **Why this is important:** BERT and similar transformer models use attention mechanisms to understand word relationships in a context-sensitive manner. For example, in a query like "Where can I fly from New York to Paris next week?" BERT could better understand the relationship between "fly," "New York," and "Paris," thereby improving location extraction and intent classification.
- **Impact on the system:** Transformer models can improve the accuracy of the system, especially when handling more nuanced or complex queries. They can also provide a deeper understanding of user intentions, which can enhance the user experience by providing more accurate travel-related responses.

2. Expansion of Source-Destination Keywords

The current system relies on a fixed list of keywords to identify the source and destination locations from user queries. However, a more dynamic and adaptive approach could be implemented to enhance its flexibility.

- **Dynamic Learning:** Continuously updating the source and destination keywords based on real user queries will help the system better handle variations in phrasing, colloquialisms, and emerging travel trends.
- **Why this is important:** Over time, people may use new terminology or phrasing that the system might not be prepared for. For instance, users may refer to new airports, cities, or popular travel routes that the chatbot is unaware of. Dynamic keyword learning can address this challenge.
- **Impact on the system:** The system would become more resilient to language changes and user behavior, allowing it to better handle a broader range of real-world queries and continuously evolve based on input from users.

3. Confidence Score Mechanism

Introducing a **confidence score** for both location detection and query classification would add another layer of reliability to the system. When the system detects a source and destination location, or when it categorizes a query, a confidence score could

indicate how certain the system is about its output.

- **Why this is important:** Not all queries will be straightforward. Some may have ambiguous or missing information. A confidence score could be used to flag uncertain results, prompting the system to ask the user for clarification. For example, if the system is not sure about the destination in the query "I want to visit a city in Italy," it could respond with, "Could you specify the city in Italy you are referring to?"
- **Impact on the system:** This mechanism would help reduce errors and improve user satisfaction by preventing the chatbot from making incorrect assumptions and improving the accuracy of responses.

4. Integration with Geolocation APIs

Integrating with **Geolocation APIs** like the **GeoText API** or **Google Places API** can further enhance the accuracy of the system's location detection. These APIs can verify the extracted locations and match them with actual cities, airports, or landmarks.

- **Why this is important:** Sometimes, a user may input ambiguous or incorrect place names (e.g., "I want to fly to Paris" when they mean Paris, Texas, not Paris, France). Using a geolocation API can validate these locations and ensure that the system is providing the correct destination.

- **Impact on the system:** This would significantly improve the reliability of location extraction and ensure that users are receiving relevant information about real destinations, avoiding potential confusion caused by ambiguous or uncommon location names.

5. Handling Complex Travel Queries

Currently, the system works well with simple queries like "book a flight from New York to Tokyo." However, more complex queries, such as those involving multi-leg journeys or ambiguous travel plans, can present challenges.

- **Why this is important:** Multi-leg journeys, such as "I am traveling from Mumbai to Delhi via Jaipur," require the system to understand the relationship between multiple destinations and legs of the trip. Similarly, users may ask for flights with specific conditions (e.g., "I want to fly from Paris to London but have a layover in Amsterdam"), which require more sophisticated query parsing.
- **Impact on the system:** Developing the system's ability to handle such complex queries will expand its use cases and improve the overall user experience by providing accurate and actionable information for more intricate travel plans.

6. Multilingual Query Support

As the chatbot system expands to a global audience, adding support for **multilingual queries** would be a crucial step in making the system more accessible and user-friendly.

- **Why this is important:** Users from different regions may speak various languages, and for a travel chatbot to be truly useful, it must accommodate queries in multiple languages (e.g., English, Spanish, French, Mandarin, etc.). This would make the system more versatile and able to handle travel-related queries from a broader user base.
- **Impact on the system:** Adding multilingual support would increase the chatbot's reach, ensuring that people from diverse linguistic backgrounds can interact with the system and receive accurate travel information in their native language.

5.2 Conclusion

This research focused on creating a travel chatbot that could accurately classify user queries, extract source and destination locations, and provide relevant travel-related

responses. Key findings from the study included:

- **Hybrid Approach for Location Detection:** A hybrid method using keyword analysis before and after location mentions yielded high accuracy for source-destination extraction. This method proved highly effective for most travel queries, making it a robust solution for identifying key travel locations.
- **Intent Classification Performance:** The performance of intent classification models varied based on the feature extraction technique used. **TF-IDF embeddings** outperformed **Word2Vec** embeddings, with TF-IDF achieving high accuracy (93.33% with Gradient Boosting). This shows that word relevance, which TF-IDF captures, is more suited for travel-related query categorization compared to Word2Vec's word embeddings, which are better at capturing word meanings in context.
- **Accuracy of the Combined Model:** When combining all the components (source-destination extraction, intent categorization, and post-processing), the system reached an overall accuracy of 75%. This indicates that the hybrid model is effective for handling the wide variety of travel queries users may have.

In summary, the research demonstrates that a combination of strong feature extraction methods, such as TF-IDF, and powerful machine learning models, such as Gradient Boosting, can result in an effective travel chatbot system. However, there are still areas for improvement, particularly in handling more complex queries and incorporating real-time feedback from users. Future enhancements like integrating transformer models, adding multilingual support, and improving the handling of multi-leg travel plans could take the system to the next level and ensure a more seamless user experience.

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